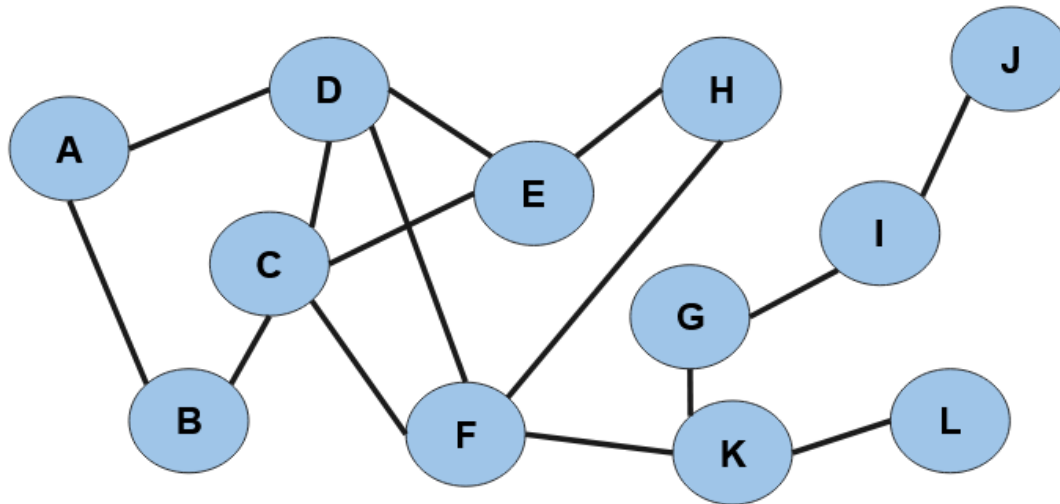


Problem Set 3

1. Answer the following questions about the below graph. Each question is 16pts.



1. Regarding connectivity, $\{V_a, V_b, V_c, V_e, V_l, V_k, V_j\}$ is a(n) _____ subgraph
2. Regarding the number of possible connections compared to existing connections in the subgraph, $\{V_c, V_d, V_f\}$ is a _____ subgraph, also known as a _____. (2pts)
3. Regarding connectivity, $\{V_a, V_b, V_c, V_d, V_e\}$ is a(n) _____ subgraph
4. The diameter of the graph is _____.
5. The degree of node C is _____.
6. Subgraph $\{V_h, V_e, V_d, V_c\}$ is an n-clique with an n of _____.
7. Subgraph $\{V_h, V_f, V_g, V_k\}$ is an n-clique with an n of _____.
8. Subgraph $\{V_a, V_b, V_c, V_d, V_e\}$ is an n-clique with an n of _____.
9. The graph is k-connected, with a k of _____.

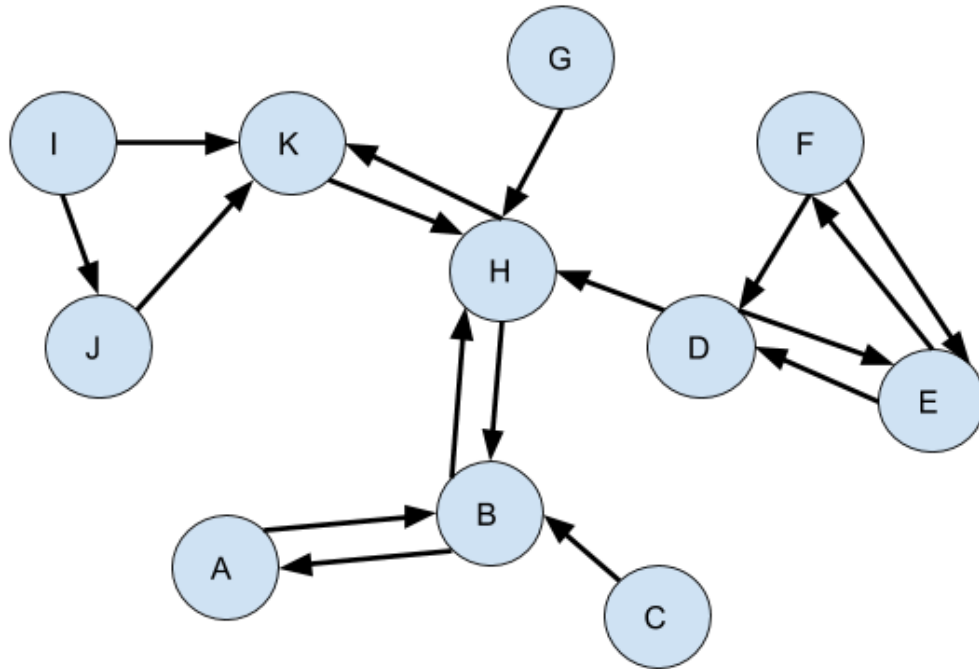
Drop node G, K, I, J, and L when considering the following questions

10. The graph is k-connected, with a k of _____.
11. Menger's Theorem predicts that the graph will have _____ independent paths between non-adjacent nodes.
12. How many independent paths connect nodes A and H?
13. How many independent paths connect nodes A and F?

14. An edge cutset of the graph is _____.

15. A UV cutset of nodes V_c and V_h is _____ (just one answer is fine).

2. Answer the following questions about the directed adjacency matrix. (9pts)



a) What is the order of the graph? (1pt)

b) What is the size of the graph? (1pt)

c) What is the maximum size of the graph? (1pt)

d) What is the average in-degree? (1pt)

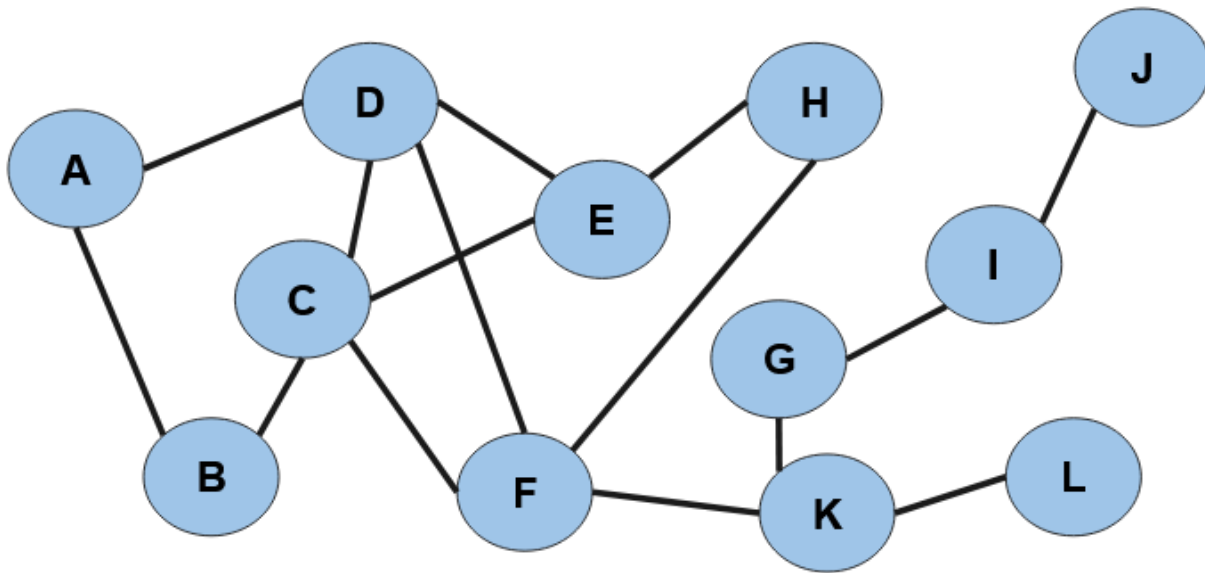
e) What is the average out-degree? (1pt)

f) What is the density of the graph? (2pts)

g) If you make a matrix from the above graph, would it be an asymmetric, symmetric, or antisymmetric matrix? (1pt)

h) What is an example of non-adjacent nodes in the graph? (1pt)

3. Answer the following questions about the undirected adjacency matrix. (8pts)



i) What is the order of the graph? (1pt)

j) What is the size of the graph? (1pt)

k) What is the maximum size of the graph? (1pt)

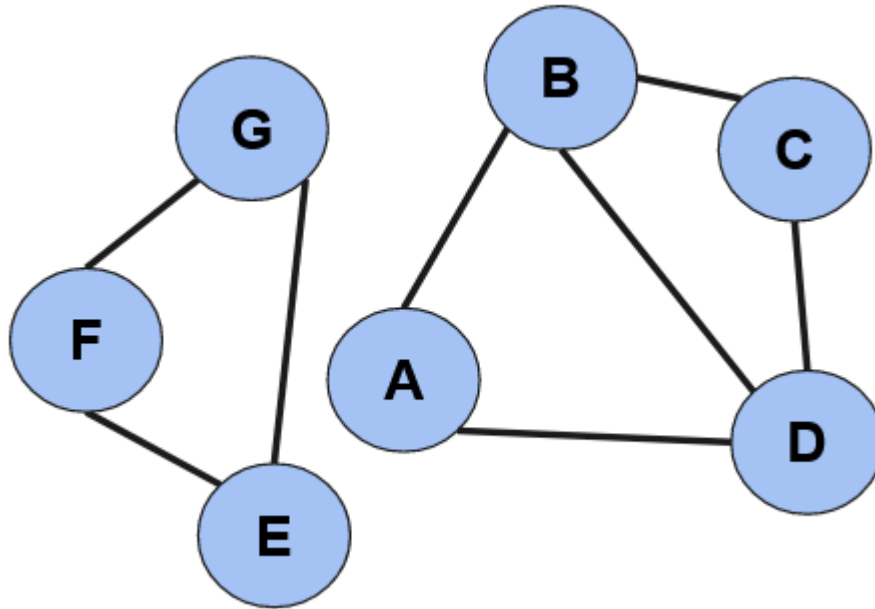
l) What is the density of the graph? (2pts)

m) If you make a matrix from the above graph, would it be an asymmetric, symmetric, or antisymmetric matrix? (1pt)

n) What is an example of non-adjacent nodes in the graph? (1pt)

o) What is the average degree in the graph? (1pt)

4. Answer the following questions about the undirected adjacency matrix. (15pts)



a. Node G has a degree of _____. (1pt)

b. What is the degree set of the above graph? (2pts)

c. What is the degree sequence of the above graph? (2pts)

d. What is the degree distribution of the above graph? (3pts)

- e. What is the density of the above graph? (2pts)
- f. What is the degree variance of the above graph? (2pts)
- g. What is the average degree in the graph? (1pt)
- h. If you make a matrix from the above graph, would it be an asymmetric, symmetric, or antisymmetric matrix? (1pt)
- i. Name a null triad in the graph (from week 1) (1pt)

6. Answer the following questions about node distributions, small worlds, and structural cohesion. (8pts)

- a. What is the difference between a scale-free network and a log-normal distribution network? (2pts) Where are “most people” in the distribution of both types of networks, and how is the expected degree of these most people different? (2 pts)
- b. What are the defining features of small world networks in terms of path length and clustering (2pts)? What does this mean for the transmission of information (1pt)?

c. Cohesive blocking uses the measure of _____ to decide which node(s) to eliminate (1pt)